

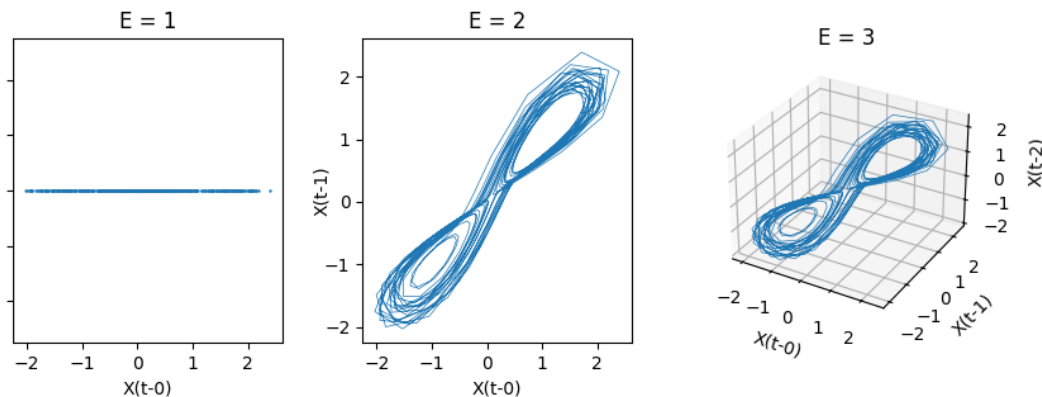
Choosing Parameters for Univariate Simplex

There are three parameters that can be adjusted when creating an embedding using a single time series: **E**, **tau**, and **Tp**.

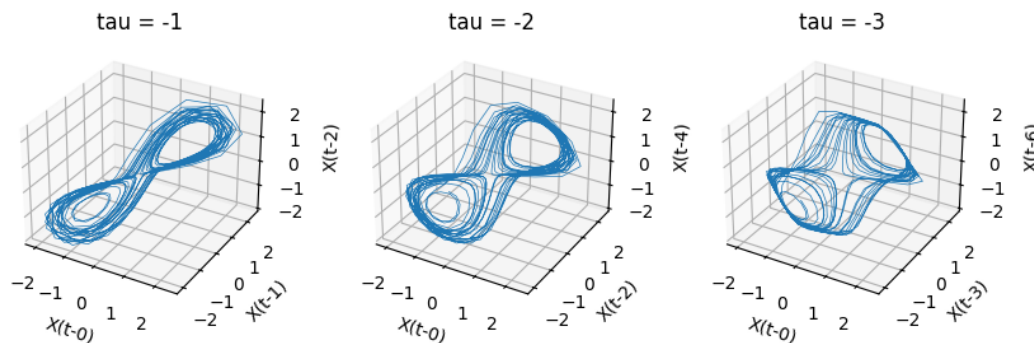
- E and tau are the **unfolding parameters** which affect how the attractor is represented in state space.
 - **E** is the embedding dimension, or number of time series lags which will be used to create the attractor reconstruction.
 - **tau** is the spacing of the time lags.

The optimal E and tau depend on the sampling frequency and system complexity.

Embeddings for X Using Different Es (tau = -1)



Embeddings for X Using Different taus (E = 3)



- **Tp** can be thought of as the **timescale** in the difference equation

$$population(t + Tp) = f(population(t))$$

The function might look linear for small Tp, especially if the data were sampled at high frequency.

To choose the best embedding parameters for the time series, we first want to set the timescale (Tp) at a fixed value and then explore E and tau.

Since we are looking to highlight the nonlinear signal while being able to create the maximum number of vectors, we want to choose the lowest E, Tp, and tau for which the Simplex algorithm provides good predictability AND does significantly better than the constant predictor (which is just the time series lagged by Tp).

- If using ρ as a metric of model performance, we use a one-tailed Fisher's z-test to determine significance.
- If using absolute error as a metric of model performance, we use a one-tailed t-test to determine significance.

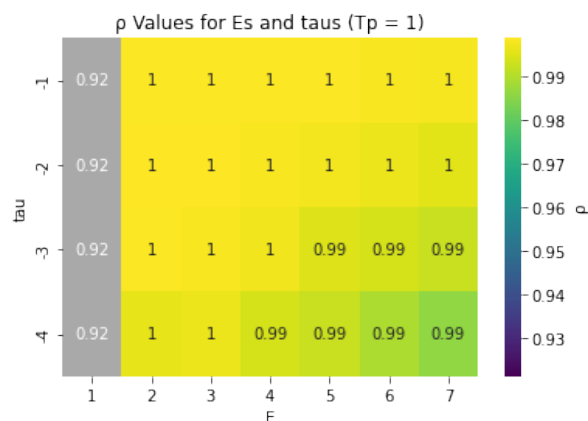
We can explore predictability for a range of E and tau values at a given Tp using the following syntax in pyEDM (default metric = 'rho', sig = 0.01):

```
target = 'X'

# Timescale
Tp = 1

# Unfolding parameters
E = [1, 2, 3, 4, 5, 6, 7]
tau = [-1, -2, -3, -4]

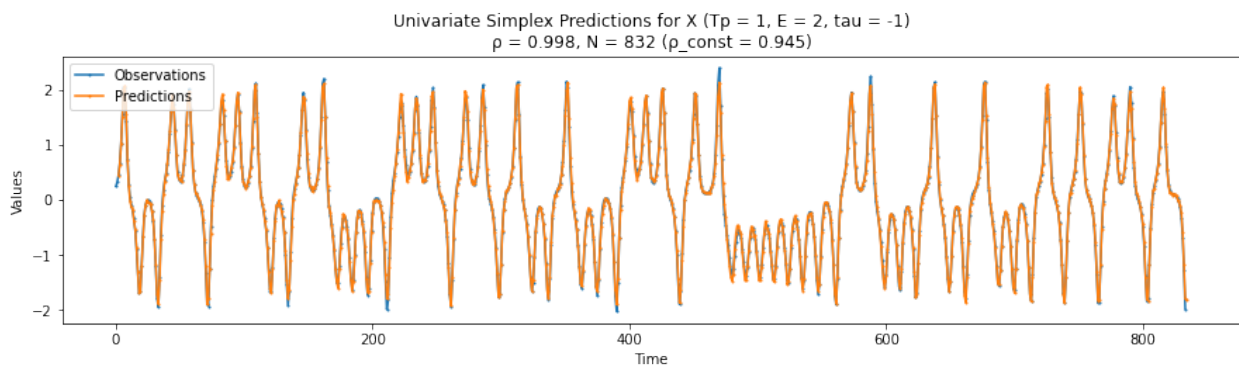
Simplex(data, target, Tp, E, tau, showPlot=True)
```



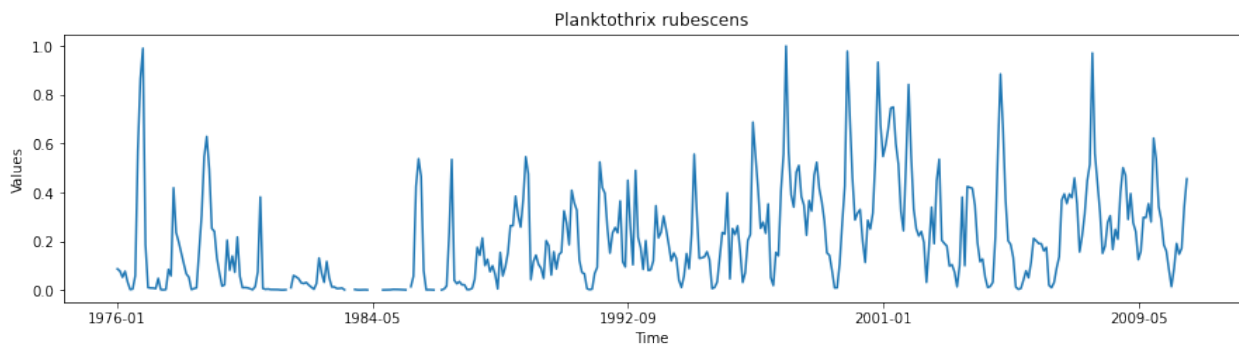
In this case, we get good predictability at $T_p = 1$, $E = 2$, $\tau = -1$. Since this is model data with no noise, we get very good predictability for many of the attractor reconstructions.

Let's look at the observed vs. predicted time series for our model data. We can do this using the following syntax in pyEDM:

```
target = 'X'  
  
# Timescale  
Tp = 1  
  
# Unfolding parameters  
E = 2  
tau = -1  
  
Simplex(data, target, Tp, E, tau, showPlot=True)
```



Let's repeat this process with observational data. *Planktothrix rubescens* is a cyanobacteria species which has been sampled monthly in Lake Zurich since the 1970s (Pomati & Shurin et al. 2020). Its time series has fairly high autocorrelation ($\rho = 0.74$).



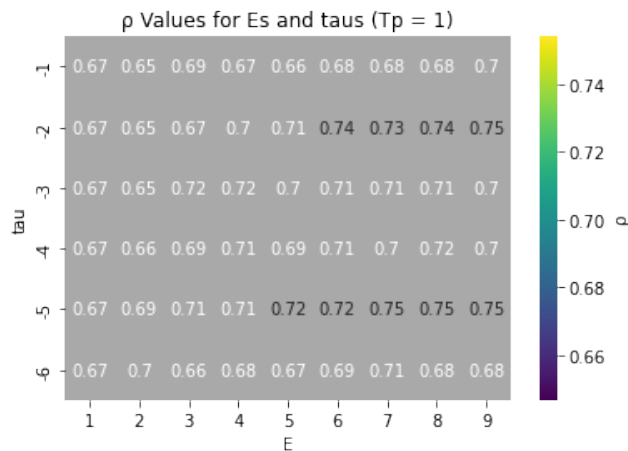
Let's start by setting the timescale at $T_p = 1$ month and exploring a range of E_s and τ values:

```
target = 'Planktothrix_rubescens'

# Timescale
Tp = 1

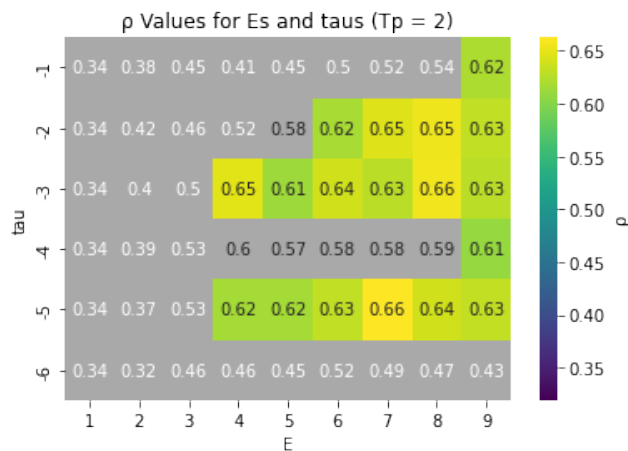
# Unfolding parameters
E = [1, 2, 3, 4, 5, 6, 7, 8, 9]
tau = [-1, -2, -3, -4, -5, -6]

Simplex(data, target, Tp, E, tau, showPlot=True)
```



The ρ values look good but no Simplex predictions do significantly better than the constant predictor on a $T_p = 1$ month timescale. Thus this timescale would not be optimal for highlighting the nonlinearity in this system.

Let's try a timescale of $T_p = 2$ months:



At this timescale, we start to detect significant nonlinear signal. We want to choose the lowest E and τ values for which we get a good and significant ρ value, which in this case would be $T_p = 2$, $E = 4$, $\tau = -3$.

Now that we have chosen our embedding parameters, let's look at the observed vs. predicted time series for *Planktothrix rubescens*:

```
target = 'Planktothrix_rubescens'

# Timescale
Tp = 2

# Unfolding parameters
E = 4
tau = -3

Simplex(data, target, Tp, E, tau, showPlot=True)
```

